

Risk Register

Scored risks, RMMM plans and a contingency reserve sized from the register

Deliverable 5 of 6 · Case Study no. 39 · PortFlow

Name	Sakshi Shingole	Roll No.	150096724056
Cohort	Jensen Huang Cohort	Course	Software Engineering & Project Management
Case	PortFlow - Container Terminal Yard and Gate Operations		

1. Method

Each risk is scored on **Probability (P)** and **Impact (I)**, both on a 1-5 scale. **Score = P × I**. Levels: 1-5 Low, 6-9 Medium, 10+ High. The scores below are the team's proposed estimates and should be reviewed with stakeholders.

2. Risk Register

ID	Risk	P	I	Score	Level
R1	Equipment integration: crane/PLC and legacy interfaces fail or arrive late	3	4	12	High
R2	Union response: resistance to slot appointments and new crane sequencing	3	3	9	Medium-High
R3	Planning engine fails or performs poorly in live operation	2	4	8	Medium
R4	Historical data is poor or incomplete for replay testing	2	3	6	Medium
R5	Port authority changes scope	2	3	6	Medium
R6	Loss of key staff	2	3	6	Medium

Probability-impact matrix

P \ I	1	2	3	4	5
5					
4					
3			R2	R1	
2			R4, R5, R6	R3	
1					

Table: red = high (10+), amber = medium (6-9), green = low (1-5)

3. RMMM Plans (Mitigation, Monitoring, Management)

Risk	Mitigation (prevent)	Monitoring (watch)	Management (if it occurs)
R1 Equipment integration	Early adapter prototype in weeks 14-16; vendor sign-off on interfaces; crane simulators	Weekly interface test pass-rate	Tablet-based manual confirmation; activate crash on D/F per Report 1
R2 Union response	Consult unions during requirements; pilot one gate first; no-job-loss guarantee on slot rules	Meeting minutes; grievance count	Phased rollout; adjust grace windows; escalate to port authority
R3 Planning engine failure	Separate execution layer; fallback rules; load tests	Plan generation time and error rate	Automatic fallback to cached plan
R4 Data quality	Data audit in weeks 1-6; cleansing scripts	% of records valid	Use a shorter clean window; supplement with simulation
R5 Scope change	Baselined SRS; change control board	Change requests per month	Impact analysis; re-baseline; draw on reserve
R6 Staff loss	Pair working; documentation; cross-training	Attrition signals; bus-factor review	Backup resource from pool

4. Contingency Reserve Sized from the Register

Rather than choosing a round percentage, the reserve is derived from the risks. Each risk's **exposure** = probability × delay (weeks) on the critical path. Mitigation lowers the probability.

Risk	Delay if it occurs (wks)	P before	Exposure before	P after	Exposure after
R1	3	0.40	1.2	0.20	0.6
R2	2	0.40	0.8	0.20	0.4
R3	1	0.20	0.2	0.10	0.1
R4	2	0.30	0.6	0.15	0.3
R5	2	0.30	0.6	0.15	0.3
R6	2	0.20	0.4	0.10	0.2
Total			3.8 wks		1.9 wks

Schedule reserve: residual exposure is about **2 weeks**, which fits inside the **3-week gap** between the 31-week expected finish and the 34-week deadline. The deadline buffer is kept rather than buying time with a crash.

Cost reserve: ₹21 lakh, the pre-approved crash of activity B (3 weeks × ₹7 L). It covers overruns up to 37 weeks, about **98.4%** of PERT outcomes (Report 1). A further ₹18 L (crash F) extends cover to 39 weeks (99.8%) and is held as management reserve, released only by sponsor approval.

5. Review Cycle

- Risk register reviewed monthly and at each milestone; scores updated using monitoring data.
- If the forecast finish exceeds 34 weeks, the crash trigger from Report 1 (B first, then F, never D) is activated.
- Closed risks are retained with the reason for closure.